

Chapter 2

AI in Air Pollution Monitoring and Its Implementation in Developing Countries

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ABSTRACT

This study looks into air pollution in developing countries and how Artificial Intelligence (AI) can contribute to air quality monitoring and data-driven interventions in those nations. Using a systematic literature review, it identifies successful AI implementations, like that from South Africa, which demonstrates cost-effective monitoring, and one from India that emphasizes the need for enabling legislation and policy. The major challenges are data integration, infrastructure constraints, and the need for energy sector reform because fossil fuel combustion remains one of the major sources of pollution. The research also suggests the potential of smart city projects to improve air pollution control. Overall, the research highlights the use of AI in addressing environmental and health issues towards the realization of Sustainable Development Goals (SDG) 3—Good Health and Well-Being, and SDG 13—Climate Action.

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Addressing Environmental Challenges With AI, Robotics, and Augmented Reality

1. INTRODUCTION

Air pollution is a serious problem for the environment and public health, especially in developing nations where urbanization, industrialization, and lax regulations all contribute to air quality decline. Conventional air pollution monitoring techniques frequently have drawbacks such as exorbitant prices, a dearth of real-time data, and insufficient coverage. On the other hand, by improving data-driven policymaking, predictive analytics, and monitoring effectiveness, artificial intelligence (AI) provides a game-changing solution.

Poor air quality remains a significant issue in many developing countries, increasing the risk of respiratory illnesses, cardiovascular complications, and early mortality. People in areas with high air pollution levels have been found to have decreased lung capacity; one study in Delhi indicated that around 40% of its population had reduced lung capacity (Rautela & Goyal, 2024). Current monitoring systems are often outdated, unavailable, or incapable of providing comprehensive and up-to-date data. AI-driven models can analyze large datasets, identify pollution trends, and make precise predictions about future air quality levels. To improve monitoring coverage and forecasting, machine learning and deep learning algorithms can utilize satellite data, Internet of Things sensors, and meteorological variables. Traditional air quality stations are expensive to build and maintain, which is not a viable solution for developing countries where people are at highest risk due to air pollution (Mckenzie et al., 2024). Here, AI-powered systems can substantially reduce monitoring costs, making them more feasible for resource-limited nations. By using cloud-based AI models to remotely access real-time air quality data, governments and regulators can make more informed decisions. Developing countries often lack the infrastructure and trained personnel necessary for extensive environmental monitoring. With minimal human involvement, AI can automate data processing and provide valuable insights. International collaboration and public-private partnerships can enhance capacity building and facilitate AI deployment across various sectors. AI-powered air pollution monitoring can equip decision-makers with evidence-based information to support the enactment of sound environmental policies. This research can help pinpoint the ethical issues, regulatory challenges, and policy frameworks needed for AI-driven environmental governance.

Research Objectives

- 1) To find out how AI can be used to mitigate the challenges of air pollution in developing countries where traditional air quality monitoring stations are expensive to build and maintain.
- 2) To find the barriers to implementing AI-based solutions for air pollution in developing countries in government policy, infrastructure, and public awareness.

Research Questions

- 1) How can AI be adapted to the unique challenges of air pollution in developing countries?
- 2) What are the barriers to implementing AI-based solutions for air pollution in these regions?

2. LITERATURE REVIEW

2.1 Air Pollution

The environment and human health are seriously threatened by air pollution, which is the release of toxic compounds into the atmosphere (Rani et al., 2011). Deforestation, automobile pollution, and industrialization are some of the causes of air pollution (Soni et al., 2020). Air pollution is toxic and causes serious respiratory and cardiovascular problems (Lawan et al., 2023). Children, the elderly, and people with pre-existing diseases are especially susceptible to the acute and long-term health impacts of both indoor and outdoor air pollution (Yamaoka, 1986). According to Rani et al. (2011), respiratory conditions, allergies, and even cancer are common health problems linked to air pollution. Primary pollutants are those that are released directly into the atmosphere, while secondary pollutants are created by chemical processes in the atmosphere (Yamaoka, 1986). Numerous control measures, including technology advancements, regulatory reforms, and the usage of cleaner energy sources, have been put into place to solve this worldwide issue (Kumar & Katoria, 2013). According to Kumar and Katoria (2013), these include enhancing fuel quality, switching to compressed natural gas for automobiles, and putting scrubbing procedures in place for industrial emissions.

2.2 Artificial Intelligence

The replication of human intellect in machines, known as artificial intelligence (AI), allows them to carry out tasks that normally demand human cognitive abilities (Nagendraswamy & Salis, 2021; Morandín-Ahuerma, 2022). Artificial intelligence can create machines which can learn, adapt and complete work that would otherwise need human intelligence (Saxena et al., 2023). Computer science, biology, psychology, and engineering are just a few of the fields that are included in artificial intelligence (Sonone & Dharme, 2019). It can be classified according to autonomy (reactive, deliberative, cognitive, or entirely autonomous) and cognitive capacity (weak, general, or superlative) (Morandín-Ahuerma, 2022). Performance in manufacturing, service systems, and expert systems has been greatly enhanced by AI applications (Pannu, 2015). These applications cover a wide range of industries, including weather forecasting, business, engineering, research, and medicine (Pannu, 2015). AI is particularly useful in the fields of computer gaming, medical picture categorization, network intrusion prevention, and power system oscillation damping (Pannu, 2015). It is anticipated that as AI technology develops, it will either complement or completely replace human abilities in a variety of domains, resulting in greater accuracy and efficiency when doing complicated tasks (Morandín-Ahuerma, 2022).

2.3 Artificial Intelligence and Environmental Management

The potential of artificial intelligence (AI) to address environmental issues and promote sustainability is increasingly recognized. Applications of AI are present in various fields, including water resource management, transportation, energy management, and biodiversity conservation (Raihan et al., 2024). These technologies enable advanced prediction models for renewable energy production while enhancing power grid management and resource allocation (Raihan et al., 2024). Furthermore, AI facilitates sustainable resource management, climate change modelling, and environmental monitoring (Shuford, 2024). Biodiversity researchers are utilizing machine learning and natural language processing techniques to forecast ecological services (Raihan et al., 2024). However, the rapid advancement of AI requires a solid legal understanding and oversight to uphold safety and ethical standards (Parveen M.A. Ansari et al., 2021). The ethical implications and potential socioeconomic impacts of AI-driven environmental solutions must be considered, even as AI offers innovative solutions for resource optimization and climate change mitigation (Lohani, 2024).

2.4 Artificial Intelligence and Air Pollution

A potent technique for predicting and forecasting air pollution is artificial intelligence (AI). With a spike in research since 2017, recent studies demonstrate the growing use of AI techniques in this field (Li et al., n.d.). Among the most popular AI techniques for predicting air quality include machine learning, deep learning, and fuzzy logic (A. Masood & K. Ahmad, 2021). Because these strategies can handle complex, non-linear data, they have demonstrated higher performance when compared to traditional statistical methods (S. Subbiah & S. Paramasivan, 2024). To anticipate the air quality index (AQI) and lower pollution on city roadways, artificial intelligence (AI)-based algorithms have been created (Rohan Gupta et al., 2024). In terms of research production, China and the US are at the top, with the Chinese Academy of Sciences holding a commanding position (Li et al., n.d.). A promising area of study for air pollution is the use of AI with environmental science techniques and inexpensive air quality sensors (Li et al., n.d.).

Research Gap

The research gap exists in the challenges of implementing AI-based solutions for air pollution mitigation in developing countries such as Bangladesh in South Asia. Thus, this research aims to find how AI can be applied to mitigate air pollution and what the challenges are in adopting AI applications.

3. METHODOLOGY

A systematic literature review methodology has been used in this work by examining academic papers, reports, and case studies of artificial intelligence (AI) in air pollution monitoring and its use in developing nations. A structured search for relevant academic literature has also been conducted across multiple databases and sources, such as academic databases, government and institutional reports, and industry and technical reports. Academic databases like Google Scholar and Semantic Scholar were used to search for resources. Most papers studied have been published in the last ten years, so the current research is thoroughly analyzed. Peer-reviewed papers from recent years were studied to ensure the authenticity of the research. Journals and case studies on AI applications for air pollution in developing countries have been analyzed to find out the different AI systems that were used, their affordability, and policy implications. Existing policies and infrastructure have also been analyzed to determine their readiness to implement AI solutions, and the

direction for future research will be identified. It was studied whether other research supported the findings from the resources used for credibility.

4. FINDINGS AND DISCUSSION

4.1 AI in Air Pollution Monitoring

4.1.1 Risks of Air Pollution

The presence of pollutants in the air from fossil fuel combustion and other industrial processes has harmful effects on human health. Exposure over a long period may cause cardiovascular disease and respiratory diseases such as asthma and bronchitis. Children and the elderly are especially prone to such risks. Cardiovascular diseases like heart attack and stroke are also caused by exposure to harmful pollutants. It can also harm mental health. Harmful pollutants also cause hypertension, headaches and lack of attention. The WHO estimates that about 4.2 million people die every year from air pollution. The risk of air pollution is particularly high in areas with high population density. In addition to human health hazards, pollutants also harm plant and animal lives, causing biodiversity loss. It also causes a lack of productivity and strain on healthcare facilities (Mckenzie et al., 2024; Rautela & Goyal, 2024).

4.1.2 Challenges with Traditional Air Pollution Monitoring

Traditional air quality and weather monitoring stations are expensive to build and maintain. Since they cannot be built easily and, in many places, there is limited coverage by traditional monitoring stations. There are also resource constraints, and the stations cannot monitor data at high frequency and cannot always do real-time monitoring (Mckenzie et al., 2024).

For these challenges, it is necessary to build cost-effective sensors that can be easily made in large quantities and provide extensive coverage for better detection of pollution sources and main hotspots. They can also provide real-time data so that more effective public health interventions can be taken and better policy decisions can be made (Mckenzie et al., 2024).

The introduction of AI and IoT has transformed air pollution monitoring by enabling real-time, data-driven decision-making. IoT-enabled sensors collect data on pollutants like PM_{2.5}, NO_x, CO, and VOCs, which AI models analyze to find patterns, identify pollution sources, and predict air quality. Unlike traditional fixed-location systems, this AI can offer greater accuracy and responsiveness. AI can generate instant pollution alerts through apps and software, allowing authorities to take prompt

action, such as issuing health advisories or rerouting traffic. In developing regions, where traditional systems are limited and expensive, low-cost IoT sensors combined with AI provide a scalable and affordable solution. This integration also supports long-term policymaking by offering insights into pollution trends and intervention outcomes, greatly enhancing public health and environmental sustainability (Oko & Rweyemamu, 2024).

Internet of Things (IoT) devices can be used to collect air pollution data, as studied by Mani et al. (2021) in their research. A prototype IOT device for collecting air pollutants was developed, which would collect air pollutants like carbon monoxide, ozone and ammonia. The device used sensor modules to register the data. Different statistical models were used to see which would give the best results in forecasting data. It was seen that the Auto-Regression Integrated Moving Average Model (ARIMA) was the better option after comparing the forecast data and test data. Though the device was used in one location, it was cost-effective to use it in different places across India for accurate real-time air pollution tracking. So, even in developing countries like India, AI can be used to monitor air pollution.

Another cost-effective solution is the use of virtual sensors for air quality monitoring, as studied by Zaidan et al. (2023). Virtual sensors are machine learning models integrated into low-cost sensors (LCS) to collect measurements of different pollutants. Such virtual sensors can be used to increase the amount of air quality monitoring data by having more monitoring stations which are small in scale. Virtual sensors can be used to monitor pollutants that are costly or difficult to measure. They do so by analyzing the correlation between pollutants and environmental variables. Thus, this allows for low-cost sensors to be deployed for wide-scale monitoring and offers a cost-effective solution for air quality monitoring.

4.1.3 Case Study SACAQM and AI_r

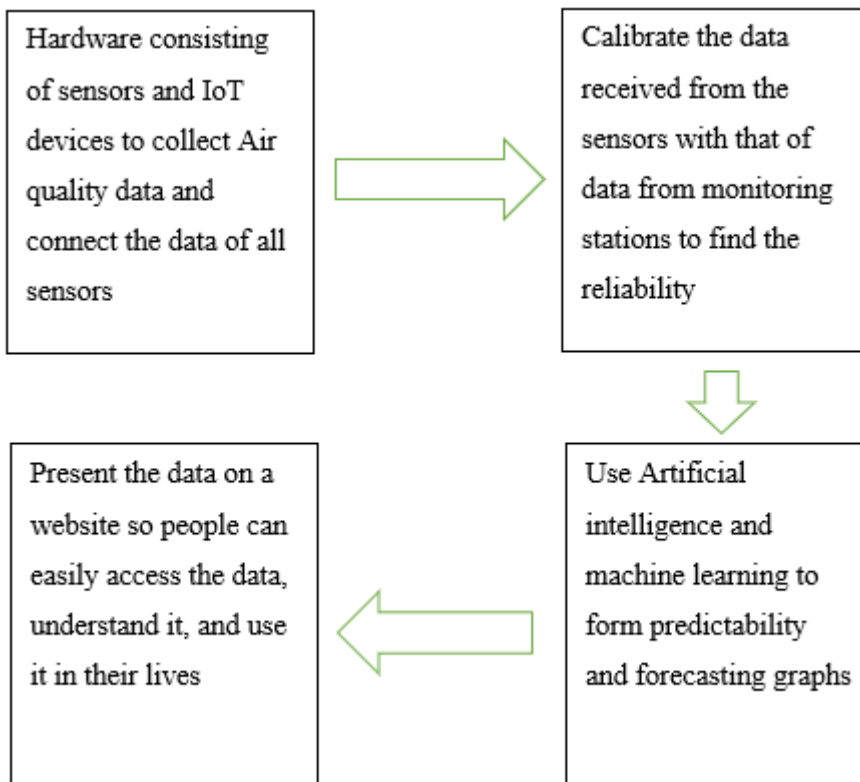
The South African Consortium of Air Quality Monitoring (SACAQM) was established with the aim of bringing together public, private and research institutions to create cost-effective air quality monitoring systems. They mainly created a system that will complement the existing air quality monitoring stations and not replace them. These will help to enhance the readings of existing stations (Mckenzie et al., 2024).

SACAQM developed the AI_r system to complement the existing monitoring stations. The AI_r consisted of a printed circuit board, which a battery or external power supply can power through a USB cable. Both ways were designed so that the system could work with very little power consumption. A Sensirion SEN55 sensor was used to monitor Particulate Matter (PM), temperature, humidity, Volatile Organic compounds (VOC), and nitrogen oxides. IoT devices were used to connect separate sensors. The data gathered from the AI_r device and traditional monitoring system

were calibrated, and it was seen that both have similar consistency and accuracy, so the AI_r system was also reliable. Calibration is used to compare the AI_r device readings and data from traditional stations and see if the AI_r data is accurate. In this regard, the comparison revealed that the system's data is reliable. The data were presented on a website where people can access it easily, understand it and make informed decisions based on the data. From the data of the nodes, Artificial Intelligence and Machine Learning were used to create predictions and forecasting for air quality data (Mckenzie et al., 2024).

The AI_r devices were deployed in school buildings in several cities in South Africa. This would raise awareness among people about air pollution and its monitoring. Thus, it would also help to see how air pollution can affect vulnerable and poor areas. Also, using real-time data, teachers can make informed decisions, like whether to continue outdoor activities for the children (Mckenzie et al., 2024).

Figure 1. Process outline of SACAQM AI_r



Note: Developed by the authors' comprehension of AI r

4.1.4 Case Study India's Encounter with Air Pollution

4.1.4.1 *Pollutants*

The main pollutants in India are particulate matter (PM), organic aerosols, black carbon, water-soluble brown carbon and volatile organic compounds. These are released from traffic emissions and residential cooking and heating. Generally, fuel combustion is the dominant source of these pollutants, so finding cleaner energy sources is very important. However, there are also atmospheric interventions, such as UV rays, that create some pollutants. Therefore, it is also necessary to gain more knowledge on air pollution (Rautela & Goyal, 2024).

4.1.4.2 *Remedy*

India has been enacting legislation aimed at tackling air pollution since 1905, and the process is still ongoing in the 21st century in the internet era. Key legislation is the Air (Prevention and Control of Air Pollution) Act of 1981, which was amended multiple times. Then, there is the Environment Protection Act 1986 and the Motor Vehicles Act. There have also been many judicial interventions and executive actions to control air pollution. Judicial interventions in the form of petitions filed by M. C. Mehta were quite influential in forming guidelines and policies. There were also executive actions, such as the Auto Fuel Policy in 2003, which was revised later to take clean energy initiatives (Rautela & Goyal, 2024).

India has been trying to monitor air quality since the 1980s when it formulated the National Ambient Air Quality Standards (NAAQS), and the Central Pollution Control Board (CPCB) has been monitoring it and providing guidelines for the public. There are Continuous Ambient Air Quality Monitoring System (CAAQMS) stations in 1082 locations for real-time data and decision-making (Rautela & Goyal, 2024).

Many studies have been conducted in India to predict air pollutants and different air quality variables using different artificial intelligence and machine learning models. The pollutants included sulfur dioxide, ozone, carbon monoxide and PM 2.5. Most studies were successful in predicting the levels of pollutants so it shows that AI models can be used for air quality monitoring in India (Rautela & Goyal, 2024).

4.1.4.3 *Challenges in India*

In the first place, there is a technological barrier as highly advanced technology is expensive, and the rural areas lack adequate infrastructure for new technological solutions. Then, there is a problem in regulation, though there are established laws, frameworks and policies that are not implemented accordingly, and thus better mechanisms to punish non-compliance are required. Thirdly, there is a large gap in public awareness about the causes and effects of air pollution, which necessitates

people to be made more aware of the hazards of air pollution so that they can take proactive measures to reduce their carbon footprints. India has many harmful agricultural practices, such as burning crop residues, which release large quantities of pollutants into the air. Farmers should be provided with proper alternatives to dispose of crop residue rather than burning it. Urbanization and infrastructure development also release pollutants such as dust and cement into the air. Sustainable and green building technologies need to be used that release fewer pollutants and prioritize air quality. Also, there is the issue of cross-border pollution, for which India needs to collaborate with its neighbours to address air quality concerns. Furthermore, since air pollution is also related to climate change, long-term climate change goals are required to cope with it. Lastly, there is also the challenge of social equity, so no group is affected more by reforms (Rautela & Goyal, 2024).

4.1.4.4 Potential for India

Advancements in Artificial Intelligence and Machine Learning models can be used with weather forecasting numerical models to predict and manage air pollution. These technologies can offer a deeper understanding of air pollution, the effect of atmospheric conditions on air pollution and the dispersal patterns of pollutants. Moreover, these technologies can create breakthroughs in sustainable energy sources and can lead the transformation from traditional fossil fuels to cleaner alternatives, thus reducing the overall pollution in India. However, India also needs to collaborate with other countries to share knowledge, find the best practices, and conduct joint research initiatives. In this way, India can contribute to global knowledge and understanding of air pollution and benefit from it. India needs access to global expertise, technologies and resources for which collaboration is essential (Rautela & Goyal, 2024).

4.2 Controlling Air Pollution by AI

Odirichukwu et al. (2024) have analyzed how AI can be used to develop early warning systems that will help people with respiratory illnesses stay inside. They used an IoT device to monitor air quality in Nigeria and find data that showed hazardous levels of air pollutants. Many IoT devices are being manufactured for use, such as the Canazree AI, but they are not available in developing countries like Nigeria, and that is an issue. Therefore, the government should put resources into building monitoring infrastructures to measure air quality. Such initiatives are required for both short-term and long-term environmental protection and achieving SDG goals 3,

6, 11, 13 and 17, which include good health, clean water and sanitation, sustainable cities, and climate action partnership for goals.

Monitoring can be done with AI, and it can be used to control air pollution in many ways. By monitoring the level of pollutants, effective policies to mitigate them can be put in place, such as industrial regulations. With AI, industrial processes can be optimized so that waste and emissions can be reduced and stopped. Further monitoring can reveal whether the industries are following the standards or not.

AI can be used in making smart grids and energy management that will help reduce loss in various systems by making them more efficient Cheng & Yu (2019). This will reduce energy waste and the demand for energy, and it will help reduce pollution caused by energy production. AI is also used to integrate various renewable energy sources into the main grid, optimize the system, and reduce costs (Boza & Evgeniou, 2021). Renewable energy sources do not pollute the environment by emitting greenhouse gases, so it can be beneficial to reduce air pollution.

AI can help in smart traffic control using real-time sensor data and forecasting to have a positive impact on mobility and the environment. AI can be integrated with GPS, surveillance cameras, and vehicles to monitor where there is a high level of traffic congestion and help people avoid those areas, thus reducing the idle time spent in traffic congestion. Besides saving time, this also means that less fuel is burnt by cars, and this means less greenhouse emissions and a reduction in air pollution (Nallaperuma, 2019).

4.3 Policy Implications

Ramadan et al. (2024) demonstrate that real-time air quality can be monitored in industrial areas for pollutants such as ammonia, carbon monoxide, nitrogen dioxide, methane, carbon dioxide, sulfur dioxide, and ozone. This research carries significant policy implications, as real-time monitoring systems facilitate continuous data collection and prompt action to address pollution levels. Additionally, such monitoring will help establish comprehensive standards for the maintenance of air quality monitoring systems, including data accuracy and system calibration. The data generated from this system will enable industries to implement measures to control emissions and develop safety protocols. Moreover, it will foster public-private partnerships, allowing collaboration to identify optimal air pollution control methods, while the government can provide assistance or subsidies to small firms addressing air pollution. Lastly, this research has implications for promoting worker safety and regulatory compliance.

4.3.1 Governance and Air Pollution Monitoring

In a study, a dataset was created on 10,000 urban centres in the Global South. The dataset examined the Air Quality Monitoring (AQM) practices in these low to middle-income cities. It was found that approximately 90% of these cities did not engage in any AQM activities, while the remaining 10% exhibited varying degrees of such activities. AQM activities were found to depend on the income levels of the cities and the characteristics of their institutions, particularly the extent of democracy present. Urban areas with higher income levels and more democratic institutions are more likely to implement AQM activities in response to high levels of air pollution. In democracies, public authorities are more responsive to the demands of citizens and facilitate the involvement of various sectors in the development and distribution of low-cost sensors. In contrast, non-democracies have fewer monitoring stations, and democracies experience a higher proliferation of low-cost sensors. Additionally, democracies conduct monitoring in both high and low-pollution areas, whereas non-democracies primarily focus on low-pollution areas (Schoch et al., 2025).

4.3.2 Implication for public health and urban planning

When compared to conventional statistical and deterministic models, AI-driven models—especially those driven by machine learning and deep learning algorithms—have shown impressive accuracy and efficiency in predicting air pollution levels. To produce accurate air quality forecasts, these sophisticated models make use of enormous volumes of historical and real-time data, such as traffic patterns, industrial emissions, satellite imaging, and meteorological variables. The ability of AI-based models to combine spatial and temporal data is one of its main advantages; this enables them to capture intricate correlations between various environmental parameters and pollution sources. Artificial intelligence (AI) systems continuously learn from data, enhancing their predicting abilities over time, in contrast to traditional models that could rely on preset equations and assumptions. Given how quickly weather patterns and human activity may alter air quality, this flexibility is especially useful for short-term air pollution forecasting (Neo et al., 2022).

Since they allow policymakers and medical practitioners to evaluate the possible health hazards associated with poor air quality, accurate air pollution forecasts are crucial for public health management. Predictive models powered by artificial intelligence (AI) can assist in predicting pollution spikes and provide early warnings, enabling communities and governments to put preventive measures in place, like limiting industrial emissions, modifying traffic patterns, or informing susceptible groups to take precautions. Healthcare providers can also use these forecasts to plan for a rise in respiratory and cardiovascular illnesses associated with exposure to air

pollution. Additionally, by incorporating air quality forecasts into environmental policy and urban planning, AI models aid in the creation of smart city projects (Neo et al., 2022).

AI-powered monitoring systems, for example, can assist in identifying the ideal sites for green areas, maximizing the use of public transit to cut emissions, and facilitating the transition to clean energy. All things considered, the use of AI in air pollution forecasting is a revolutionary step toward proactive environmental management and the defence of public health. AI-driven algorithms help to create healthier, more sustainable urban settings by consistently enhancing the precision and dependability of air quality predictions (Neo et al., 2022).

4.4 International and Regional Policies on AI in Environmental Management

4.4.1 EU

The European Union (EU) is making major efforts to incorporate sustainability considerations into its AI legislation to guarantee that AI technologies support environmental objectives while respecting fundamental rights and ethical norms. The EU is investigating methods to reinterpret current legislative frameworks, like the General Data Protection Regulation (GDPR), in a way that is consistent with sustainability goals as part of this endeavour. For instance, the GDPR's data-sharing rules might be reexamined to allow AI-powered emissions tracking and environmental monitoring while upholding stringent privacy safeguards. The EU is proposing new laws, most notably the EU AI Act, which intends to create comprehensive guidelines for the creation, application, and use of AI systems, in addition to amending existing ones. The AI Act places more stringent rules on high-risk AI systems and classifies AI applications according to risk levels. Additionally, the law promotes accountability, transparency, and moral AI practices, with sustainability emerging as a crucial factor. To ensure that AI systems help lower energy usage and carbon footprints, the EU is anticipated to incorporate environmental impact evaluations into AI regulatory compliance (Perucica & Andjelkovic, 2022; Hacker, 2022).

The EU is adopting co-regulation—a governance model that blends industry-driven standards and best practices with regulatory oversight—to achieve a balance between individual rights and larger environmental concerns. This strategy makes it possible for AI development to be flexible while maintaining environmental sustainability as a top concern. The EU wants to establish an AI ecosystem that links technical advancement with climate action and sustainable resource management by working with companies, researchers, and civil society. Sustainability by design, which stresses incorporating environmental factors into AI development

from the beginning, is another essential element of the EU's approach. These imply that resource optimization, carbon reduction, and energy efficiency should all be considered when designing AI systems. For example, AI-driven data centres could have to run on renewable energy, and AI models might be tuned to use the least amount of computing power possible.

With these diverse initiatives, the EU hopes to establish itself as a world leader in sustainable AI governance and establish a standard for how AI can be governed to benefit the environment and society. It is anticipated that incorporating sustainability concepts into AI regulations will have an impact on other areas and further the worldwide goal of developing AI in an ecologically conscious manner (Perucica & Andjelkovic, 2022; Hacker, 2022).

4.4.2 China

Thanks to robust government support and technology breakthroughs, Chinese businesses are increasingly using artificial intelligence (AI) to improve environmental management. The Chinese government has acknowledged AI as a crucial instrument for accomplishing its lofty sustainability objectives, which include becoming carbon neutral by 2060. China is aggressively pushing companies to include AI-driven solutions for energy efficiency, pollution control, and sustainable resource management through funding, legislation, and regulatory frameworks (Sun et al., 2024).

Energy optimization is one of the main uses of AI in environmental management. Smart grids driven by AI assist in balancing the supply and demand for electricity, cutting down on energy waste and improving power distribution efficiency. AI algorithms are being used by many Chinese businesses, especially in sectors like manufacturing, logistics, and construction, to forecast patterns in energy usage, modify operations accordingly, and lower their carbon footprints. AI-driven predictive maintenance reduces energy loss and equipment downtime in industrial facilities by ensuring that machinery runs as efficiently as possible (Sun et al., 2024).

AI is also being used to check the quality of the air and water, which is important considering China's history of pollution-related issues. AI-powered sensor networks are being deployed by businesses to continually monitor air pollution, identify dangerous pollutants, and give regulatory agencies real-time information. AI is being utilized in the water industry to optimize wastewater treatment, guaranteeing that contaminants are effectively eliminated before water is released back into the environment.

Furthermore, AI is revolutionizing the creation of smart cities. To cut emissions and increase sustainability, cities like Beijing, Shanghai, and Shenzhen are incorporating AI into trash management, transportation, and urban planning. Artificial intelligence (AI)-powered traffic management systems maximize road utilization by

minimizing traffic and fuel consumption, which immediately improves air quality (Sun et al., 2024).

The success of these AI-powered environmental projects emphasizes the importance of public-private partnerships in China. To encourage companies to work with academic institutions and government organizations to create AI-based green solutions, the government has started several programs, such as the AI Innovation Action Plan for Ecological and Environmental Protection. Furthermore, big private companies like Alibaba, Baidu, and Tencent, as well as state-owned enterprises (SOEs), are aggressively funding AI-driven sustainability initiatives. China is establishing itself as a global leader in AI-driven environmental management by encouraging cooperation between the public and commercial sectors, which proves that AI can be a potent force for industrial sustainability and ecological preservation. Chinese businesses will be vital in determining how AI-powered environmental governance develops in the future as these technologies advance (Sun et al., 2024).

4.5 Global Energy Sector

Artificial intelligence (AI) is changing the global oil and gas sector by transforming environmental management, promoting sustainability, and supporting the transition to cleaner energy. Countries like the USA, UK, Sweden, and the UAE are using AI to optimize drilling and extraction, waste reduction, and boost energy efficiency through tools like machine learning, predictive maintenance, and digital twins. AI also assists in detecting and mitigating environmental hazards such as carbon emissions, methane leaks, and oil spills via drones, IoT sensors, and satellites. Plus, AI supports the integration of renewable energy in the energy grids by balancing outputs from solar and wind with traditional oil and gas infrastructure. It also assists in regulatory compliance, helping companies monitor emissions and forecast policy shifts. These innovations are advancing sustainability goals while modernizing energy systems globally (Chutcheva et al., 2022).

4.6 AI, Smart Cities and Air Pollution Control

United Nations Sustainable Development Goal 11 is to create sustainable cities and communities that place importance on economic opportunities, safety, resilience, and sustainability. It also includes robust infrastructure with public transportation and green spaces, plus less resource consumption and less waste and emission. Smart cities focus on using technology to improve efficiency and provide intelligent solutions for energy efficiency, waste management and other criteria. So, smart cities and sustainable cities complement each other; that is, smart cities can become

sustainable. Global South has been seeing a rapid increase in urbanization, with many city centres aiming to become smart cities (Das, 2025).

AI can be used in many different sectors in a smart city, where the main sectors are planning and management and the economy include finance and optimum resource allocation. Then, there is infrastructure and mobility that consists of transportation and mobility, energy management and waste management. AI can also help in environmental monitoring and reducing emissions that will help reduce air pollution. It will also help in governance, participation, responsiveness, equity, quality of life, healthcare and education (Das, 2025).

4.6.1 Barriers to Smart Cities

There are many barriers to creating smart, sustainable cities; first of all, there is an infrastructure deficit as many cities do not have good broadband internet networks, reliable energy grids, and efficient transport systems. Also, many cities have unequal access to healthcare and education. Then, effective policies for digital technology and artificial intelligence are not in place in many cities. Plus, many cities have limited access to funding and investment.

4.6.2 Smart Cities in the Global South

Despite the barriers, many cities are implementing smart city initiatives. First of all, India has created smart financial solutions in digital payment scenarios, which increase transparency and accountability and reduce corruption. Kenya is also implementing smart city initiatives in main city centres like Nairobi, which have smart grid technologies enabled by AI that monitor real-time energy consumption and help reduce waste. Rwanda also has a digital payment system and has implemented an IoT-enabled car parking system to reduce congestion (Das, 2025). Thus, smart city initiatives can be used to monitor and enhance energy usage and traffic so that there will be less wastage and more efficient energy consumption, leading to better air quality as less fuel will be needed for energy.

4.6.3 Policy Recommendations for Implementing Smart Cities

- Cities should identify key areas that require technological innovation, such as transportation, healthcare, waste management and energy infrastructure and prioritize these sectors first.
- City authorities should encourage partnerships between the public and private sectors so that the private sector has funds to research, implement changes

and involve international experts and other stakeholders, which will increase efficiency and transparency.

- Governments should develop their institutional capacities through training in AI and collaboration with universities and research institutes.
- Encourage people's participation so that marginalized communities will not be left behind, and create inclusive policies so that all people will benefit equally and the digital divide will be reduced.

4.7 Key Challenges in AI Adoption and Solutions

Artificial Intelligence (AI) adoption is a game-changer, especially in fields like public services, healthcare, and environmental management. However, a number of significant obstacles stand in the way of its optimal application and broad adoption. To guarantee that the advantages of AI are optimized, these issues need to be resolved through cooperation, efficient governance, and ethical concerns.

4.7.1 Data Integration Issues

Data integration is one of the biggest obstacles to the adoption of AI. For AI systems to make wise conclusions, high-quality, varied, and structured data is essential. Integrating data from various sources, including government databases, Internet of Things sensors, satellite imaging, and public records, can be challenging for both public and private companies. Additionally, the data could be inaccurate, fractured, or compartmentalized, which makes training and using AI models more difficult (Uwaoma et al., 2024). Establishing interoperable data systems that can easily communicate and process information across several platforms and stakeholders is necessary to overcome data integration challenges. Businesses and governments must invest in data processing and cleaning technology, establish open data rules, and standardize data formats.

4.7.2 Lack of AI Literacy

The lack of AI literacy among important stakeholders, such as legislators, corporate executives, and the broader labour force, is another significant obstacle. To make wise decisions in the complicated realm of artificial intelligence, one must have a firm grasp of algorithms, data analysis, and machine learning concepts. There are skill gaps that restrict people's capacity to use and profit from AI technologies in many developing nations and even in some developed economies (Uwaoma et al., 2024). Governments and educational establishments should establish AI education and training initiatives at all levels, from professional development centres to univer-

sities, to address this. Decision-makers should be the focus of AI literacy programs to make sure they can oversee AI initiatives and comprehend their ramifications.

4.7.3 Resistance to Technological Change

Many facets of society, including both persons and institutions, frequently oppose technological advancement. People can worry that AI would require large expenditures in new infrastructure, cause data privacy issues, or replace jobs. A societal hesitancy to embrace AI is frequently also a result of doubts about its efficacy or its unforeseen repercussions (Uwaoma et al., 2024). Clear communication about the advantages of AI and its potential long-term benefits—such as improved productivity, cost savings, and the creation of jobs in new industries—is necessary to overcome reluctance. Governments can take a proactive approach to change management by offering incentives for the adoption of AI and developing frameworks that facilitate workforce changes, such as retraining initiatives. AI solutions should also be made easy to use and integrated into current systems to lessen resistance and friction. As was seen from the case study in India, people are still clinging to old practices of burning crop residue, which greatly increases the level of pollutants in the air. People are less aware of the causes and consequences of air pollution, and thus, they need to be educated about such practices which damage the environment. Governments should develop other sustainable methods of waste disposal rather than burning, which are less harmful to the environment, and AI can be used to make the new systems efficient.

4.7.4 Ethical Considerations

Several ethical questions, including those about privacy, bias, accountability, and transparency, may be brought up by AI technologies. For example, if the data used to train AI algorithms reflects past disparities, the algorithms may unintentionally reinforce biases. Furthermore, because AI systems frequently depend on vast amounts of sensitive or personal data to operate efficiently, worries about data privacy are intensified. It is also challenging for people and organizations to comprehend how and why decisions are made when AI decision-making systems are opaque (Uwaoma et al., 2024). Strong regulatory frameworks must be created to guarantee that AI systems are developed and implemented ethically and responsibly to lessen these ethical dilemmas. Organizations and businesses must also embrace ethical AI standards that support the eradication of prejudice and protect individuals' privacy.

4.7.5 Collaborative Efforts and Robust Governance

Lastly, cooperation amongst various stakeholders—including governments, businesses in the private sector, academic institutions, and civil society—is necessary to overcome the obstacles to AI adoption. Addressing the cross-cutting problems related to AI deployment requires a concerted effort. To ensure that AI advances the public benefit without violating people's rights or escalating social injustices, strong governance frameworks must be established to supervise its development and application. Public forums should be formed for people's participation in AI and air quality monitoring issues (Uwaoma et al., 2024). As was seen from the study by Schoch et al. (2025), it is important for democratic participation to ensure that there is adequate air quality monitoring. So, general people also need to be more active in voicing their problems due to air pollution and pressure the authorities to act in the public's interests.

4.8 Future Research Implication

This study paves the way for future research that can address more local challenges, such as the consequences of limited infrastructure, scarcity of data for AI language models, and policy gaps in local governance. Future studies can work on solutions that can be implemented in a more local and decentralized way that are affordable and scalable and cater to local needs. Low-cost sensors and AI models should be given more importance for air quality monitoring and the use of data for public health and environmental policies. Additionally, inclusive and sustainable policy frameworks that encourage community understanding and participation need to be given due importance.

5. CONCLUSION

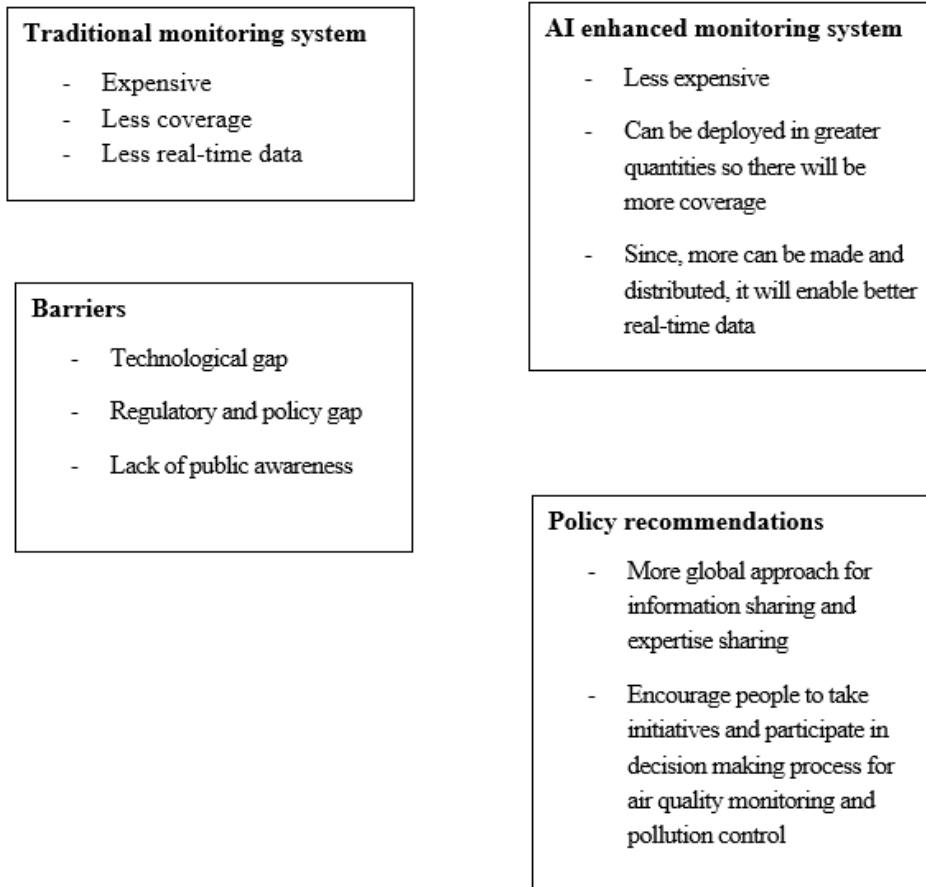
AI has the potential to transform air pollution monitoring, particularly in underdeveloped nations where conventional methods often fail due to a lack of infrastructure, resources, and data. Developing countries can enhance their ability to predict pollution levels, monitor air quality in real time, and implement prompt interventions by leveraging AI technologies such as machine learning, predictive modelling, and IoT integration. In addition to offering a more economical and effective way to evaluate and reduce air pollution, AI-driven solutions can be tailored to address specific local challenges, thereby enhancing environmental preservation, public health, and policymaking. As demonstrated in the earlier discussion, devel-

oping countries like India and Nigeria have successfully experimented with AI data monitoring. Therefore, it can be applied globally to mitigate air pollution.

However, several challenges must be addressed for AI to be effectively utilized in air pollution monitoring in underdeveloped nations, including poor data integration, inadequate technical infrastructure, and low levels of AI literacy. Strong governance structures, adequate training for local stakeholders, and ethical considerations are also essential for the implementation of AI technology to ensure equitable access and prevent increasing social inequality.

Governments, international organizations, and the business sector must collaborate and invest in infrastructure development, capacity building, and the establishment of strong regulations to ensure the long-term effectiveness of AI in air pollution monitoring. With proper support, AI has the potential to significantly improve air quality control, enhance citizen well-being, and assist developing nations in achieving their environmental and public health goals. Undoubtedly, digital technology will play a crucial role in the future of air pollution monitoring, with AI serving as a key enabler of this transformation.

Figure 2. Summary of findings compiled by the authors



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